SHEET 4

1. Consider the following program fragment:

```
int a [10];
int i;
for(i = 0; i<10; i++)
    a[i] = i*i;

for(i=0;i<10;i+=2)
    cout << a[i];
```

What will this program fragment display?

2. Write code segment to read 10 floats and print
    a. The maximum
    b. The minimum
    c. The Average

9. Mean and standard deviation. Write a program that reads a set of floating-point data values from the input. When the end of file is reached, print out the count of the values, the average, and the standard deviation. The average of a data set {x1, ..., xn} is , where is the sum of the input values. The standard deviation is

$$s = \sqrt{\frac{\sum \left(x_i - \bar{x}\right)^2}{n-1}}$$

However, this formula is not suitable for the task. By the time the program has computed, the individual xi are long gone. Until you know how to save these values, use the numerically less stable formula

$$s = \sqrt{\frac{\sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2}{n-1}}$$

You can compute this quantity by keeping track of the count, the sum, and the sum of squares as you process the input values.

4 -

6) Write a program that returns number of occurrences of a string in another string. E.g.

Input:

First String: the

Second String: **the** students are working hard in **the** faculty of Engineering because **the**y love it

Output:

3

Note: You should use nested loops. Don't use some methods like indexOf or substring

5 –

Write a program that's take a 15 integer number and make sort for it at ascending and descending order and print it in ascending and descending .

6 –

1) Write a program that prints the reverse of a given string (using loops).

Input:

CODE

Output:

EDOC